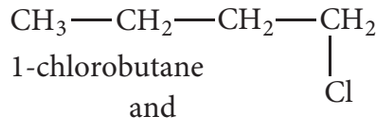
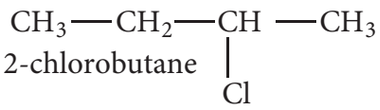
(ii) Mol. formula C_4H_9Cl



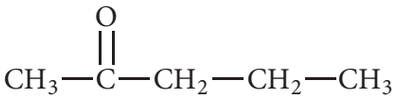
1-chlorobutane

and



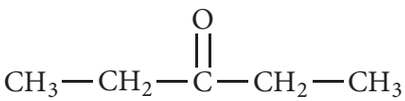
2-chlorobutane

(iii) Mol. formula $C_5H_{10}O$



Pent-2-one

and



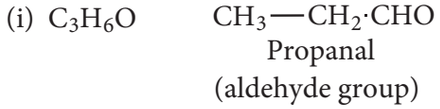
Pent-3-one

(c) Functional isomerism:

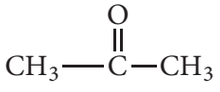
Different compounds having same molecular formula but different functional groups are said to exhibit functional isomerism.

Example:

(i) C_3H_6O

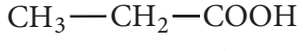


Propanal
(aldehyde group)

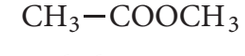


Propanone
(keto group)

(ii) $C_3H_6O_2$



Propanoic acid
acid group



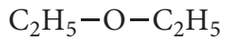
Methyl acetate
(ester group)

(d) Metamerism: This type of isomerism is a special kind of structural isomerism arises due to the unequal distribution of carbon atoms on either side of the functional group or different alkyl groups attached to the either side of the same functional group and having same molecular formula. This isomerism is shown by compounds having functional group such as ethers, ketones, esters and secondary amines between two alkyl groups.

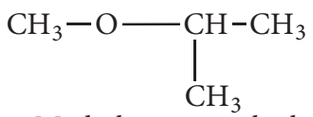
(i) $C_4H_{10}O$



Methyl propyl ether
1-methoxypropane

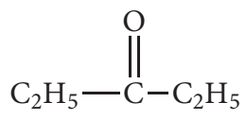


diethyl ether
ethoxyethane

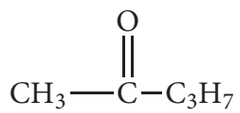


Methyl iso-propyl ether
2-methoxypropane

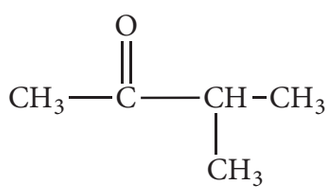
(ii) $C_5H_{10}O$



diethyl ketone
pentan-3-one



Methyl propyl ketone
pentan-2-one



Methyl isopropyl ketone
3-methylbutan-2-one

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